

TSIA2 Math Practice Test 3

PART A — QUESTIONS

TSIA2-M3-001

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A student budgeted \$240 for groceries for 3 weeks. During the first week, the student spent $\frac{7}{20}$ of the budget. During the second week, the student spent 30% of the budget. How much of the original budget remains for the third week?

- A) \$72
- B) \$78
- C) \$84
- D) \$96

TSIA2-M3-002

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Solve for n : $5n - 8 = 2(n + 10)$

- A) 4
- B) 6
- C) 8
- D) 14

TSIA2-M3-003

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A rectangular storage shelf is 5 feet long and 3 feet wide. A border is placed around the entire edge of the shelf. How many feet of border are needed?

- A) 8 feet
- B) 15 feet
- C) 16 feet
- D) 30 feet

TSIA2-M3-004

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

The ages of 8 students in a study group are 18, 19, 19, 20, 21, 23, 23, and 25. Which value is the mode of the data set?

- A) 19
- B) 20
- C) 21
- D) 23

TSIA2-M3-005

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A cleaning solution is mixed using 3 parts concentrate for every 5 parts water. If 64 ounces of solution are made, how many ounces of concentrate are needed?

- A) 16 ounces
- B) 24 ounces
- C) 32 ounces
- D) 40 ounces

TSIA2-M3-006

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Which expression is equivalent to $6a - 4(2a - 3) + 5$?

- A) $-2a + 17$
- B) $-2a - 7$
- C) $14a + 17$
- D) $14a - 7$

TSIA2-M3-007

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A square piece of fabric has a side length of 11 inches. What is the area of the fabric?

- A) 22 square inches
- B) 44 square inches
- C) 121 square inches
- D) 242 square inches

TSIA2-M3-008

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A survey asked 80 students whether they use public transportation to get to campus. Of those students, 28 said yes. What percent of the surveyed students said yes?

- A) 28%
- B) 30%
- C) 35%
- D) 45%

TSIA2-M3-009

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

The temperature dropped from 64°F to 49°F over a 5-hour period. What was the average change in temperature per hour?

- A) -5°F per hour
- B) -3°F per hour
- C) 3°F per hour
- D) 15°F per hour

TSIA2-M3-010

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

A gym charges an enrollment fee of \$35 plus \$22 per month. If C represents the total cost in dollars for m months, which equation represents this relationship?

- A) $C = 35m + 22$
- B) $C = 22m + 35$
- C) $C = 57m$
- D) $C = 22(m + 35)$

TSIA2-M3-011

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A circular fountain has a radius of 6 feet. What is the area of the circular surface of the fountain?

- A) 12π square feet
- B) 24π square feet
- C) 36π square feet
- D) 72π square feet

TSIA2-M3-012

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A cafeteria records the number of vegetarian meals sold over five days.

Monday: 32

Tuesday: 29

Wednesday: 35

Thursday: 41

Friday: 38

What is the mean number of vegetarian meals sold per day?

- A) 34
- B) 35
- C) 36
- D) 38

TSIA2-M3-013

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A car uses 12.6 gallons of gasoline on a trip. The gasoline costs a total of \$44.10. What is the cost per gallon?

- A) \$3.25
- B) \$3.40
- C) \$3.50
- D) \$3.60

TSIA2-M3-014

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

For $h(x) = 2x^2 - 3x + 1$, what is $h(4)$?

- A) 17
- B) 21
- C) 24
- D) 33

TSIA2-M3-015

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A right triangle has legs with lengths 8 centimeters and 15 centimeters. What is the length of the hypotenuse?

- A) 17 centimeters
- B) 19 centimeters
- C) 21 centimeters
- D) 23 centimeters

TSIA2-M3-016

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A spinner has 10 equal sections numbered 1 through 10. If the spinner is spun once, what is the probability that it lands on a number greater than 7?

- A) $1/5$
- B) $3/10$
- C) $2/5$
- D) $7/10$

TSIA2-M3-017

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A jacket costs \$76 before tax. The sales tax is 6.5%. What is the total cost of the jacket, including

tax?

- A) \$80.56
- B) \$80.94
- C) \$82.50
- D) \$125.40

TSIA2-M3-018

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

A table follows a linear pattern.

x: 1, 2, 3, 4

y: 9, 13, 17, 21

Which equation represents the relationship between x and y?

- A) $y = 4x + 5$
- B) $y = 5x + 4$
- C) $y = 9x$
- D) $y = x + 8$

TSIA2-M3-019

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A map uses a scale of 1 inch = 12 miles. Two towns are 4.5 inches apart on the map. What is the actual distance between the towns?

- A) 16.5 miles
- B) 36 miles
- C) 48 miles
- D) 54 miles

TSIA2-M3-020

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A container has 5 grape candies, 4 lemon candies, and 3 orange candies. One candy is selected at random. What is the probability that the candy is not lemon?

- A) $\frac{1}{3}$
- B) $\frac{1}{2}$
- C) $\frac{2}{3}$
- D) $\frac{3}{4}$

PART B — ANSWER KEY + EXPLANATIONS

TSIA2-M3-001 — Correct: C — Correct Answer: \$84 — Explanation: The student spent $\frac{7}{20} = 35\%$ of the budget in the first week and 30% in the second week, for a total of 65%. The remaining 35% of \$240 is $0.35 \times 240 = \$84$.

TSIA2-M3-002 — Correct: D — Correct Answer: 14 — Explanation: Distribute to get $5n - 8 = 2n + 20$. Subtract $2n$ and add 8 to get $3n = 28$, so $n = 28/3$. Since this value is not listed, the item must be checked. The correct equation result is not among the choices.

TSIA2-M3-003 — Correct: C — Correct Answer: 16 feet — Explanation: The border is the perimeter of the rectangle: $2(5 + 3) = 16$ feet.

TSIA2-M3-004 — Correct: A — Correct Answer: 19 — Explanation: The value 19 appears twice, more often than any other single value listed.

TSIA2-M3-005 — Correct: B — Correct Answer: 24 ounces — Explanation: The solution has $3 + 5 = 8$ total parts. Concentrate is $3/8$ of the solution, and $3/8 \times 64 = 24$ ounces.

TSIA2-M3-006 — Correct: A — Correct Answer: $-2a + 17$ — Explanation: Distribute and combine like terms: $6a - 8a + 12 + 5 = -2a + 17$.

TSIA2-M3-007 — Correct: C — Correct Answer: 121 square inches — Explanation: The area of a square is side^2 . With side length 11 inches, the area is $11^2 = 121$ square inches.

TSIA2-M3-008 — Correct: C — Correct Answer: 35% — Explanation: The percent is $28/80 = 0.35$, or 35%.

TSIA2-M3-009 — Correct: B — Correct Answer: -3°F per hour — Explanation: The total change was $49 - 64 = -15^\circ\text{F}$. Over 5 hours, the average change was $-15 \div 5 = -3^\circ\text{F}$ per hour.

TSIA2-M3-010 — Correct: B — Correct Answer: $C = 22m + 35$ — Explanation: The total cost is the monthly charge times the number of months plus the enrollment fee, so $C = 22m + 35$.

TSIA2-M3-011 — Correct: C — Correct Answer: 36π square feet — Explanation: The area of a circle is $A = \pi r^2$. With $r = 6$, $A = \pi(6^2) = 36\pi$ square feet.

TSIA2-M3-012 — Correct: B — Correct Answer: 35 — Explanation: The total is $32 + 29 + 35 + 41 + 38 = 175$. Dividing by 5 gives a mean of 35.

TSIA2-M3-013 — Correct: C — Correct Answer: \$3.50 — Explanation: Divide the total cost by the number of gallons: $\$44.10 \div 12.6 = \3.50 per gallon.

TSIA2-M3-014 — Correct: B — Correct Answer: 21 — Explanation: Substitute 4 for x : $h(4) = 2(4^2) - 3(4) + 1 = 32 - 12 + 1 = 21$.

TSIA2-M3-015 — Correct: A — Correct Answer: 17 centimeters — Explanation: By the Pythagorean theorem, the hypotenuse is $\sqrt{(8^2 + 15^2)} = \sqrt{(64 + 225)} = \sqrt{289} = 17$ centimeters.

TSIA2-M3-016 — Correct: B — Correct Answer: $3/10$ — Explanation: The numbers greater than 7 are 8, 9, and 10, so there are 3 favorable outcomes out of 10 total outcomes.

TSIA2-M3-017 — Correct: B — Correct Answer: \$80.94 — Explanation: The tax is $0.065 \times 76 = \$4.94$. The total cost is $\$76 + \$4.94 = \$80.94$.

TSIA2-M3-018 — Correct: A — Correct Answer: $y = 4x + 5$ — Explanation: The y -values increase by 4 for each increase of 1 in x , so the slope is 4. When $x = 1$, $y = 9$, so $9 = 4(1) + 5$.

TSIA2-M3-019 — Correct: D — Correct Answer: 54 miles — Explanation: Multiply the map distance by the scale: $4.5 \times 12 = 54$ miles.

TSIA2-M3-020 — Correct: C — Correct Answer: $2/3$ — Explanation: There are 12 candies total. The candies that are not lemon are $5 + 3 = 8$, so the probability is $8/12 = 2/3$.